#114, 7th Cross, Bogadi II Stage, Mysuru - 570026, Karnataka, India, Ph: 0821-2548600, E-mail: amritamysuru@amrita.edu

COURSE PLAN

Department	MCA
Course Title	Computer Vision
Course Code	24CSA529
Programme and section	MCA
Academic Year	2026-27
Semester	III
LTP/ hours per week- Credit	L: 03, T: 00 and P: 01, 4 Credits
Course Instructor Details	Mr. Suresha R sureshar@my.amrita.edu +91 9741324649

Programme Outcomes:

PO1	**Computational Knowledge:** Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
PO2	**Problem Analysis:** Identity, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
PO3	**Design /Development of Solutions:** Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	**Conduct Investigations of Complex Computing Problems:** Use research-based knowledge and research methods, including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern Tool Usage:** Create, select, adapt, and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.

PO6	**Professional Ethics:** Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practice.
PO7	**Life-long Learning:** Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional
PO8	**Project management and finance:** Demonstrate knowledge and understanding of the computing and management principles and apply these to one’s work, as a member and leader in a team, to manage projects and in multidisciplinary environments
PO9	**Communication Efficacy:** Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
PO10	**Societal and Environmental Concern:** Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts and the consequential responsibilities relevant to professional computing practice
PO11	**Individual and Teamwork:** Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments
PO12	**Innovation and Entrepreneurship**: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large

Course Outcomes (COs):

CO1	Understand the fundamentals of computer vision, their applications, image types and basic operations
CO2	Apply various image processing techniques to binary images and understand different color spaces
CO3	Understand image representation using different feature extraction algorithms
CO4	Understand the methods used for analysis of videos

Prerequisites:
1. Basic programming knowledge
2. Basic mathematics

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Lesson Plan

Week	Lec. No.	Topic	Key Concepts	Teaching Aids	Textbook	COs
1	01	Introduction to Computer Vision	Introduction of Computer Vision	Presentation & Blackboard	TB1 **Ch. 1**	CO1
	02	Image Processing vs Computer Vision	Differences between Image Processing and Computer Vision	Presentation & Blackboard	TB1 **Ch. 1**	CO1
	03	Applications of Computer Vision	Object Detection and Recognition Applications	Presentation & Blackboard	TB1 **Ch. 1**	CO1
2	04	Types of Images	Binary, Grayscale, Color Images	Presentation & Blackboard	TB1 **Ch. 1**	CO1
	05	Image Channels	Splitting and Merging Channels, Manipulating Color Pixels	Presentation & Blackboard	TB1 **Ch. 1**	CO1
	06	Mathematical Operations on Images	Linear and Non-Linear Operations	Presentation & Blackboard	TB1 **Ch. 3**	CO1
3	07	Mathematical Operations on Images	Arithmetic and Logical Operations	Presentation & Blackboard	TB1 **Ch. 3**	CO1
	08	Binary Image Processing	Thresholding	Presentation & Blackboard	TB1 **Ch. 3**	CO2
	09	Morphological Operations	Erosion and Dilation	Presentation & Blackboard	TB1 **Ch. 3**	CO2
4	10	**Class Test – I**				
	11	Morphological Operations	Opening and Closing	Presentation & Blackboard	TB1 **Ch. 3**	CO2
	12	Connected Component Analysis	4-Connectivity, 8-Connectivity, Diagonal Connectivity, M-Connectivity	Presentation & Blackboard	TB1 **Ch. 3**	CO2
5	13	Contour Analysis	Boundary Detection	Presentation & Blackboard	TB1 **Ch. 3**	CO2

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	14	Color Spaces	RGB, HSV	Presentation & Blackboard	TB2 **Ch. 3**	CO2
	15	Color Spaces	CMYK, Y'CbCr, Y'UV	Presentation & Blackboard	TB2 **Ch. 3**	CO2
6	16	Image Filtering	Spatial and Frequency Domain Filters	Presentation & Blackboard	TB1 **Ch. 3**	CO2
	17	Smoothing	Mean, Max, Min, Weighted Average Filters	Presentation & Blackboard	TB1 **Ch. 3**	CO2
	18	Gradient	Gradient Vectors of an Image	Presentation & Blackboard	TB2 **Ch. 5**	CO2
7	19	Feature Extraction	Introduction to Feature Extraction	Presentation & Blackboard	TB1 **Ch. 4**	CO3
	20	Edge Features	Canny Edge Detection	Presentation & Blackboard	TB1 **Ch. 4**	CO3
	21	Edge Features	LOG and DOG Edge Detection	Presentation & Blackboard	TB2 **Ch. 5**	CO3
Midterm Exam						
8	22	Line Detectors	Hough Transform for Line Detection	Presentation & Blackboard	TB1 **Ch. 4**	CO3
	23	Corner Detection	Harris and Hessian Affine Corner Detection	Presentation & Blackboard	TB2 **Ch. 5**	CO3
	24	Orientation Histogram	Orientation Angle Frequency Feature Extraction	Presentation & Blackboard	TB2 **Ch. 5**	CO3
9	25	SIFT	Scale-Invariant Feature Transform	Presentation & Blackboard	TB2 **Ch. 5**	CO3
	26	SURF	Speeded Up Robust Features	Presentation & Blackboard	TB1 **Ch. 4**	CO3

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	27	HOG	Histogram of Oriented Gradients	Presentation & Blackboard	TB2 **Ch. 5**	CO3
10	28	GLOH	Gradient Location and Orientation Histogram	Presentation & Blackboard	TB2 **Ch. 5**	CO3
	29	Scale-Space Analysis	Image Pyramids	Presentation & Blackboard	TB2 **Ch. 4**	CO3
	30	Scale-Space Analysis	Gaussian Derivative Filters	Presentation & Blackboard	TB2 **Ch. 4**	CO3
11	31	Gabor Filters	Local Feature Analysis using Gabor Filters	Presentation & Blackboard	TB1 **Ch. 3**	CO3
	32	DWT	Discrete Wavelet Transform	Presentation & Blackboard	TB1 **Ch. 3**	CO3
	33	LBP	Local Binary Pattern	Presentation & Blackboard	TB2 **Ch. 5**	CO3
12	34	GLCM	Gray Level Co-occurrence Matrix	Presentation & Blackboard	TB2	CO3
	35	Video Analysis	Motion Estimation using Optical Flow	Presentation & Blackboard	TB1 **Ch. 8**	CO4
	36	Video Stabilization	Camera Motion Estimation and Correction	Presentation & Blackboard	TB1 **Ch. 8**	CO4
13	37	Object Tracking	Tracking Movement of Objects	Presentation & Blackboard	TB2 **Ch. 11**	CO4
	38	Kalman Filter	Object Position Estimation and Tracking	Presentation & Blackboard	TB2 **Ch. 11**	CO4
	39	MeanShift and CamShift	Fixed and Adaptive Search Window Tracking	Presentation & Blackboard	TB2 **Ch. 11**	CO4

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14	40	Background Subtraction and Modelling	Foreground and Background Detection	Presentation & Blackboard	TB2 **Ch. 11**	CO4
	41	Optical Flow	Image Motion Description	Presentation & Blackboard	TB2 **Ch. 11**	CO4
	42	KLT Tracker	Kanade–Lucas–Tomasi Feature Tracker	Presentation & Blackboard	TB3 **Ch. 9**	CO4
15	43	Spatio-Temporal Analysis	Analysis of Spatial and Temporal Data	Presentation & Blackboard	TB3 **Ch. 9**	CO4
	44	Dynamic Stereo	Event-Based Stereo Vision and Intensity Change Detection	Presentation & Blackboard	TB3 **Ch. 9**	CO4
	45	Motion Parameter Estimation	Pel-Recursive Algorithm (PRA) and Block-Matching Algorithm (BMA)	Presentation & Blackboard	TB3 **Ch. 9**	CO4
End Semester Exam						

Lab Sessions

Lab No.	Experiment / Topic	Key Concepts	Textbook & Chapter	CO Mapping
01	Introduction to Image Processing and Computer Vision	Introduction to Image Processing, Computer Vision, Basic Python Packages with Examples	**TB1 – Ch. 1**	CO1
02	Types of Images	Binary, Grayscale, Color Images, Splitting and Merging Image Channels	**TB1 – Ch. 2**	CO1
03	Mathematical Operations on Images	Arithmetic, Logical, Linear, and Non-Linear Operations on Images	**TB1 – Ch. 2**	CO1
04	Connectivity and Morphological Operations	Image Connectivity, Erosion, Dilation, Opening, and Closing	**TB1 – Ch. 3**	CO2
05	Color Space Models	RGB, HSV, CMYK, Y'CbCr, and Y'UV Color Spaces	**TB1 – Ch. 4**	CO2
06	Image Filtering	Smoothing Filters and Gradient Filters	**TB1 – Ch. 5**	CO2

07	**Lab Test – I**			
08	Feature Extraction	Canny Edge Detection, LOG, and DOG Features	**TB2 – Ch. 1**	CO3
09	Corner Feature Extraction	Harris Corner Detection and Hessian Affine Detection	**TB2 – Ch. 2**	CO3
10	Feature Extraction	SIFT, SURF, and HOG Feature Extraction	**TB2 – Ch. 3**	CO3
11	Scale-Space Analysis	Mean Pyramids and Gaussian Pyramids	**TB2 – Ch. 4**	CO3
12	Local Feature Analysis	Gabor Filters and Discrete Wavelet Transform (DWT)	**TB2 – Ch. 5**	CO3
13	Shape and Texture Features	Local Binary Pattern (LBP) and Gray Level Co-occurrence Matrix (GLCM)	**TB2 – Ch. 6**	CO3
14	**Lab Test – II**			
15	Object Tracking using Kalman Filter	Track and Estimate the Position of Objects in Video Sequences	**TB3 – Ch. 2**	CO4

CO-PO Mapping

PO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	1	-	-	-	1	-	-	-	-	-
CO2	2	2	2	1	1	-	1	-	-	-	-	-
CO3	2	2	2	2	2	-	2	-	-	-	-	-
CO4	2	2	2	2	2	-	2		-	-	-	1

Session by Alumni/ Industry Expert:

1. Session on Computer Vision

Textbooks/ References

1. Computer Vision: Algorithms and Applications by Richard Szeliski
2. Computer Vision: A Modern Approach (Second Edition) by David Forsyth and Jean Ponce

3. Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Second Edition, Cambridge University Press

Internal Assessment Component

Sl. No.	Component Name	Weightage	CO Mapping	Tentative date
1	CT 01	05	CO1	4th week
2	Midterm Examination	20	CO1, CO2, CO3	8th week
3	Lab Test 1	15	CO1, CO2	7th week
4	Lab Test 2	15	CO3, CO4	14th week
5	Project (Implementation + Documentation)	15 (10 + 5)	CO1, CO2, CO3, CO4	15th week

Name & Signatures

1. Course Teachers:

2. Class Representatives:

3. Class Advisors:

4. Program Coordinator:

5. Chairperson: